

Meerkat: A Social Community Support Application for Older Adults

Tamir Mendel
Tel Aviv University
tamirmendel@mail.tau.ac.il

Eran Toch
Tel Aviv University
erant@tauex.tau.ac.il

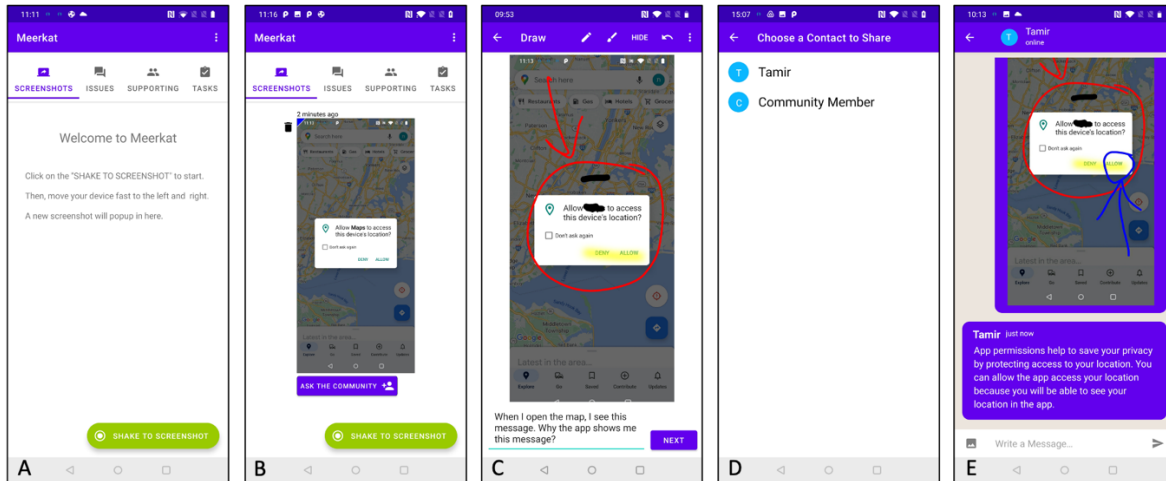


Figure 1: Screenshots of the Meerkat app. The support process contains several steps from the viewpoint of a user who request support. (A) the user takes a screenshot of the interaction of the phone by clicking the “shake to screenshot” and then shaking the device to take a screenshot. (B) Based on the screenshot, the user can ask for support from a community member. (C) the user can draw on the screenshot and write a message to describe and highlight the problem. (D) the seeker can select support and receive support from a user she knows or a user she does not know. (E) the seeker can see the helper’s message, doodle on the overlay of the screenshot, and communicate with the seeker.

ABSTRACT

Meerkat is a social support app that helps older adults tackle hurdles in interacting with mobile applications by receiving contextual support from friends and family or community volunteers. The app allows users to describe the interaction issue, capture a screenshot, draw, and write queries to helpers who can provide understandable explanations. Meerkat is informed by theories of collective efficacy and aims to reduce the barriers to existing social networks. Given COVID-19 social distancing measures, the design of Meerkat adapts patterns of peer support to situations where the users are not co-located. To users who do not have available social support, the app provides a foundation to receive support from a community of volunteers who are social connections to other users. Meerkat also includes a practice mode, facilitating social learning by presenting users with mobile tasks. We conclude with how Meerkat can contribute to collective and individual efficacy.

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CCS CONCEPTS

• **Human-centered computing** → Ubiquitous and mobile computing; Ubiquitous and mobile devices; Smartphones; Collaborative and social computing; Collaborative and social computing systems and tools; Social networking sites; • **Social and professional topics** → User characteristics; Age; Seniors.

KEYWORDS

Peer support, social support, help, mobile application, older adults

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1 INTRODUCTION

Mobile technology has become highly imperative in almost any aspect of people’s contemporary living, such as social, work, health, and education. The smartphone ownership rate in 2021 among Americans aged 18 and above was 85%, while among American people aged 65 and above was 61% [15]. Older adults experience more difficulties controlling their mobile devices [11]. With the

development of new mobile technologies and increases in life expectancy, mobile devices barriers may become more severe for older adults. One way to safely deliver mobile devices for older adults is by providing accessible and available support [6]. Social support can increase older adults' ability to handle digital technology and teaches them how to use it [3, 5, 8]. Older adults often turn to family members or friends for support and advice [10, 12]. Mendel and Toch have shown that young adults are highly willing to assist their older relatives with online security and privacy [7]. However, older adults may feel uncomfortable asking for support from close connections or do not have available helpers. Our application provides a community of helpers available to all users.

In this paper, we present Meerkat, a mobile application for Android, allowing users to receive support about mobile issues from the helpers. Users can have multiple roles; they can be seekers, users who receive technical support, or helpers who are users that provide technical assistance. When users want to receive support, they can capture a screenshot and ask another user for explanations and advice (see Figure 1). Mendel et al. have shown that overlaying screenshots with explanations can help guide older adults in the support process [8]. The user can receive support from two types of people: close contacts (e.g., family and friends) and a community of volunteers who are social connections of other seekers. Figure 2 demonstrates what the seeker can select from the helpers' lists. The seeker may choose family and friends or prefer volunteers. Seekers may choose the community of volunteers when the social networks are unavailable or feel less comfortable receiving support from their social connections. The seeker can choose the helper and receive screenshots with overlays and explanations. Our application is designed for older adults and appropriate to materialize to any other seeker segments that may need support, such as users with low self-efficacy.

Previous studies presented applications that support improving older adults' security and privacy decisions. AppMoD is an example of an application that allows older adults to delegate security and privacy decisions to a trusted social connection, such as children or grandchildren [13]. Aljallad et al. presented an application that helps users collaborate with close people to decide on application permissions [1]. EverTutor used interactive tutorials on devices and showed that overlays tutorials are particularly suited for older users when older participants had equal or faster completion times using interactive tutorials than younger participants [14]. However, these applications do not use a community of volunteers that teach seekers through the support process. The existing social connections and community of volunteers manifest the theory of collective efficacy. Collective efficacy is defined as the group's shared perception of its ability to successfully perform some behavior together [2]. Available support may increase collective efficacy and the seeker's ability to address mobile interaction problems successfully. For example, collective efficacy may help older adults face challenges when other users support them in fixing security and privacy problems [9].

Another way to take advantage of available support is through a practice mode that allows users to learn about mobile issues before they happen on their devices. The practice mode in our application shows different mobile tasks, and older adults can ask for support from the helpers. Prior studies documented age-related challenges with technology adoption, such as a perceived decline in cognitive

abilities, lack of social opportunities to interact with peers, and barriers dealing with changing technologies [4]. In our application, we demonstrate practice mode in mobile safety situations. This way, we can increase older adults' awareness, learnability and refine the memory with support from the helpers.

With Meerkat, we offer the following contributions:

Support process allows users to share and get feedback on a screenshot using overlays to notate support requests and explanations (see Figure 1).

Community of helpers allows users to receive support regarding mobile device issues. The community includes helpers who make the support available. Users can receive support from their social contacts or the social connections of other users (see Figure 2).

Practice mode provides the users' tasks to learn from them by receiving support from the helpers. The seeker can receive support in advance. For example, practice mode can teach users and increase awareness about security attacks.

2 MEERKAT

Meerkat is an Android application that allows users to receive support about mobile interaction issues from the community. It also provides a practice mode feature that presents hypothetical scenarios before they happen in the user devices to learn from helpers. Meerkat implements the support process, community, and practice mode.

2.1 Support process

Meerkat offers several features that make the support process more usable. Figure 1 shows an example of a support process when an application requests permission. In this example, the support process starts when the user clicks on "shake to screenshot" to add a screenshot to the Meerkat app by shaking the device (A). We offer users to capture a screen and share it, which without the application, the process is currently complicated for execution. A new screenshot was added to the **screenshot tab**, and the user can "ask the community" (B), then the user can write text in the box below and draw on the screenshot with a yellow marker, a red pen, or hide private information with a black brush. For example, the user can hide the application's name that requests permission (see Figure 1 C). The seeker can select users from two types of people: close relationships (e.g., family and friends) and a community of volunteers who are social connections of other seekers (D). After the user sends the message, the helper can write back a short explanation and draw on the same screenshot the seeker sent, which is useful when the helper and the seeker may have different operating systems (E). The seeker can find the messages on the **issues tab**, including conversation threads with the helpers. The helper can find the messages on the **supporting tab**, including conversation threads with the seekers. When users click on a specific conversation, they can see the text and image messages exchanged.

2.2 Community of helpers

The community of helpers refers to the users available to provide support. The seeker can be connected to the users who have social contacts (e.g., close friends, family members) or a community of volunteers who are social connections of other users who want to

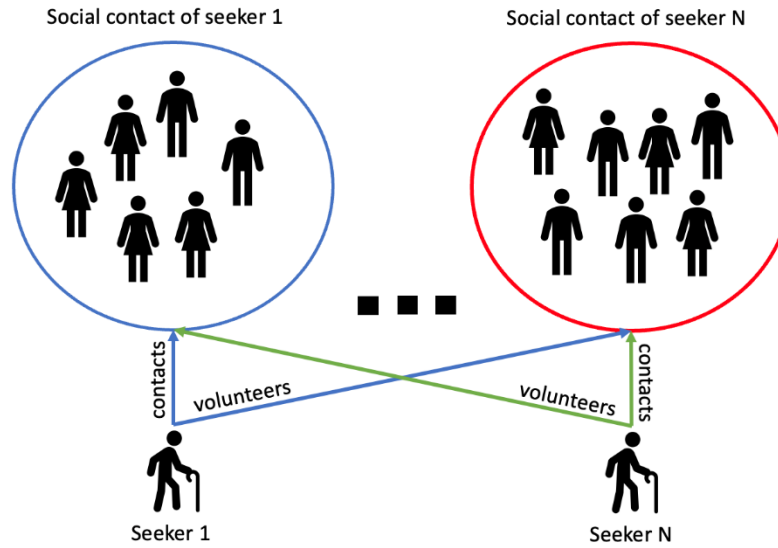


Figure 2: users can receive support from their social contacts or social connections of other users.

provide support. Figure 2 demonstrates what the seeker can select from the helpers' lists. Older adults may prefer the community when the social support networks are not available or feel uncomfortable asking their social connections for support. Figure 1 D shows a small example of a helpers' community, one of the helpers is presented with a name imported from the contact list of the user device. It implies that there is a social connection between the seeker and helper. The second helper is called a "community member," there is no social connection between the seeker and the community member. Future work can explore the allocation algorithm that offers suitable helpers to specific seekers based on availability and type of the problem.

2.3 Practice mode

Meerkat provides a practice mode feature that presents tasks on the user device to learn from helpers. When the users click on the menu item "Practice mode," Meerkat starts to post scenarios to users periodically. The task presents scenarios that can appear in the user device, and the users can ask helpers for advice and explanations about the hypothetical scenario. The **tasks tab** contains the task from the system (see Figure 3). When the user clicks on the task, the system first provides general information. When the user clicks "start," a potential issue is displayed. For example, the system displays phishing attack messages. Users can click on "take a screenshot," draw on it and share it with a helper for explanation and advice.

3 CONCLUSION

Meerkat is a social support application that allows users to describe a mobile issue and write queries to helpers. The helper can provide explanations and advice in an understandable language and teach through support. The application enables users to capture a screenshot, draw and write queries, and send the message to the helper. Our application contains two types of helpers: close relationships

(e.g., family and friends) and a community of volunteers who are social connections of other users who want to provide support. Older adults may prefer the community when social support is unavailable or feel less comfortable requesting support from their social connections. We also present the practice mode feature that shows different mobile tasks, and users can ask for support from the community of helpers. The users can ask the helper and receive support in advance.

We conclude that Meerkat can contribute to collective and individual efficacy. The community of helpers manifests the collective efficacy theory and provides available support that potentially increases collective efficacy. Support can contribute to users' learning and may improve their computer self-efficacy.

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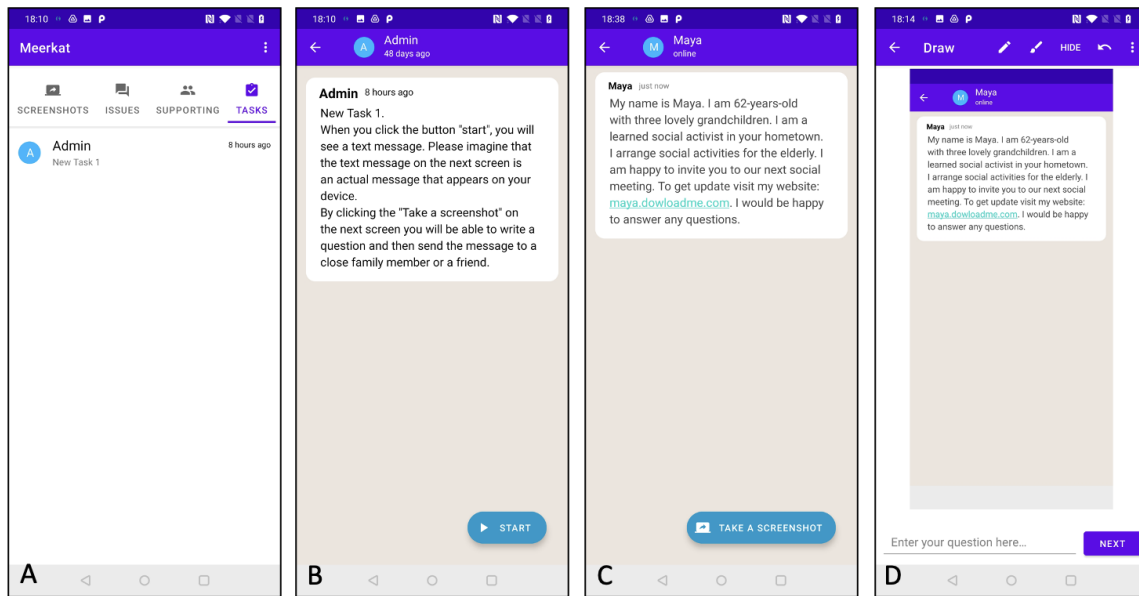


Figure 3: Practice mode example. A) the system sent a message to the user. B) The user can read general information about the task. C) The user can see a phishing example and “take a screenshot.” D) the user can draw on the screenshot and write a message to share with the community.

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